FEDERAL SPECIFICATION

SEWING MACHINES, INDUSTRIAL, FLATBED, SINGLE-NEEDLE

(DETAIL SPECIFICATION)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification 00-S-256.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for single needle, flatbed, industrial sewing machines.

1.2 Classification.

1.2.1 Styles and classes. The sewing machines shall be of the following styles and classes, as specified (see 6.2).

Style A - Stitch type 301

Class 2 - Light and medium duty
Class 3 - Medium-heavy duty
Class 4 - Heavy duty
Class 6 - Extra-heavy duty

Style C - Stitch type 101

Class 1 - Light duty basting edge
Class 3 - Basting edge, with edge worker

2. APPLICABLE DOCUMENTS

Latest issue of Federal Specification:

00-S-256 - Sewing Machines, Industrial (General Specification)

FSC 3530

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3. REQUIREMENTS

3.1 Style A. The machine shall be capable of producing stitch type 301 and shall come equipped with bobbin winder.

3.1.1 Class 2, light and medium duty. The machine shall be suitable for stitching on light or medium weight fabrics (see 6.1).

3.1.1 Feed mechanism, sewing speed, stitch regulation, and working space. The feed mechanism, sewing speed, stitch regulation, and working space shall conform to the requirements specified in table I. The kind of feed and working space shall be as specified (see 6.2).

TABLE I. Feed mechanism, sewing speed, stitch regulation, and working space, class 2

Feed mechanism (00-S-256)	Sewing speed 1/ (stitches per minute)	Stitch regulation (stitches per inch)	Working space 2/ at right of needle (inches)
Feed 1	4,800	8-18	8 or 10 to 13
Feed 4	3,800	6-16	1-1/2
Feed 7	3,500	9-16	10-1/2
Feed 8	2,900	8-18	5-3/4 or 10-1/2
Feed 16	5,000	7-18	1-1/2
Feed 17	4,500	7-18	3

1/ Top operating speed of not less than shown.

2/ Tolerance of plus 1 inch and minus 1/2 inch.

3.1.1.2 Mounting. The machine mounting shall be fully submerged in a recessed cutout in the table top. The machine shall be stand mounted.

3.1.1.3 Lubrication. Machines equipped with feed 8, compound feed, shall be nonautomatically lubricated. All other machines having various other feed mechanisms shall be automatically lubricated.

3.1.2 Class 3, medium-heavy duty. The machine shall be suitable for general tailor shop use (see 6.1).

3.1.2.1 Feed mechanism, sewing speed, stitch regulation, and working space. The feed mechanism, sewing speed, stitch regulation and working space shall conform to the requirements specified in table II. The kind of feed shall be as specified (see 6.2). When specified (see 6.2), a reversible feed mechanism shall be furnished.

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3.1.2.2 Mounting. The requirements specified in 3.1.1.2 shall apply.

TABLE II. Feed mechanism, sewing speed, stitch regulation, and working space, class 3

Feed mechanism (00-S-256)	Sewing speed 1/ (stitches per minute)	Stitch regulation (stitches per inch)	Working space 2/ at right of needle (inches)
Feed 1	2,000	6-20	10 to 13
Feed 2	1,800	6-20	10-1/4

1/ Top operating speed of not less than shown.

2/ Tolerance of plus 1 inch and minus 1/2 inch.

3.1.2.3 Lubrication. The machine shall be nonautomatically lubricated.

3.1.3 Class 4, heavy duty. The machine shall be suitable for stitching light and medium canvas goods, and articles made of heavyweight fabrics (see 6.1).

3.1.3.1 Feed mechanism, sewing speed, stitch regulation, and working space. The feed mechanism, sewing speed, stitch regulation, and working space shall conform to requirements specified in table III. The kind of feed shall be as specified (see 6.2).

TABLE III. Feed mechanism, sewing speed, stitch regulation, and working space, class 4

Feed mechanism (00-S-256)	Sewing speed 1/ (stitches per minute)	Stitch regulation (stitches per inch)	Working space 2/ at right of needle (inches)
Feed 4	3,800	6-16	3
Feed 7	4,000	6-12	10
Feed 8	2,200	6-12	10
Feed 17	4,500	7-18	3

1/ Top operating speed of not less than shown.

2/ Tolerance of plus 1 inch and minus 1/2 inch.

3.1.3.2 Mounting and lubrication. The requirements specified in 3.1.1.2 and 3.1.1.3 shall apply.

3.1.4 Class 6, extra heavy duty. The machine shall be suitable for stitching extra heavy fabric (see 6.1).

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3.1.4.1 Feed mechanism, sewing speed, stitch regulation, and working space. The feed mechanism, sewing speed, stitch regulation, and working space shall conform to the requirements specified in table IV. The kind of feed and working space shall be as specified (see 6.2). When specified (see 6.2), a reversible feed mechanism shall be furnished.

TABLE IV. Feed mechanism, sewing speed, stitch regulation, and working space, class 6

Feed mechanism (00-S-256)	Sewing speed 1/ (stitches per minute)	Stitch regulation (stitches per inch)	Working space 2/ at right of needle (inches)
Feed 2	400	2-1/2-8	15 to 18-1/2
Feed 8	1,100	2-1/2-8	20 to 30

1/ Top operating speed of not less than shown.

2/ Tolerance of plus or minus 1/2 inch.

3.1.4.2 Table top. Unless otherwise specified (see 6.2), the table top for feed 2, drop feed machines shall be not less than 30 inches wide by 58 inches long; for feed 8, compound feed machines, the top shall be not less than 32 inches wide by 42 inches long for a machine having a working space of 14 inches, not less than 54 inches long for a machine having a working space of 20 inches, and not less than 66 inches long for a machine having a working space of 30 inches.

3.1.4.3 Mounting. The requirements specified in 3.1.1.2 shall apply.

3.1.4.4 Lubrication. The machine shall be nonautomatically lubricated.

3.1.4.5 Light fixture. The light fixture for the machines having a number 2 feed mechanism shall be mounted on the sewing machine head.

3.2 Style C. The machine shall be capable of producing stitch type 101.

3.2.1 Class 1, light duty, basting edge. The machine shall be suitable for basting, and joining pieces of goods together, and other operations on light, medium, or heavy weight fabrics (see 6.1).

3.2.1.1 Feed mechanism, sewing speed, stitch regulation, and working space. The feed mechanism, sewing speed, stitch range, and working space shall conform to the requirements specified in table V. The working space shall be as specified (see 6.2).

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TABLE V. Feed mechanism, sewing speed, stitch regulation, and working space, class 1

Feed mechanism (00-S-256)	Sewing speed 2/ (stitches per minute)	Stitch regulation (stitches per inch)	Working space 3/ at right of needle (inches)
Feed 1	3,500 1/ 4,000	4-8	5 or 11

1/ For nonautomatic lubrication only.

2/ Top operating speed of not less than shown.

3/ Tolerance of plus 1 inch and minus 1/2 inch.

3.2.1.2 Mounting. The machine mounting shall be fully submerged.

3.2.1.3 Lubrication. The machine shall be either nonautomatically or automatically lubricated.

3.2.2 Class 3, basting edge, with edge worker. The machine shall be suitable for basting and other operations on light, medium, or heavy weight fabrics (see 6.1).

3.2.2.1 Edge worker. The machine shall be equipped with a power operated device for working out the edge and basting it at the same time, using only one drive belt.

3.2.2.2 Sewing speed. The machine shall have a top operating speed of not less than 1,200 stitches per minute.

3.2.2.3 Feed mechanism. The machine shall be equipped with feed 1, drop feed, mechanism, with the exception that a spring may be used to assist the lowering action of the feed.

3.2.2.4 Stitch regulation. A stitch regulator shall be provided, capable of adjusting the length of stitch over a range from 2 to 4 stitches per inch.

3.2.2.5 Working space. A clear working space to the right of the needle is not essential.

3.2.2.6 Mounting. The machine mounting shall be nonsubmerged.

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3.2.2.7 Lubrication. The machine shall be nonautomatically lubricated.

3.2.2.8 Drive. An electric transmitter or a separate motor not less than 1/4 hp, and a sleeve or ball bearing clutch and brake arrangement, mounted on a common base, shall be provided to drive the machine.

4. QUALITY ASSURANCE PROVISIONS

(See 00-S-256)

5. PREPARATION FOR DELIVERY

(See 00-S-256)

6. NOTES

6.1 Intended use.

6.1.1 Style A, class 2 machines shall be suitable for stitching flags, guidons, chevrons, light- and medium-weight shirts, dresses, coats, trousers, work clothing, blankets, suits, overcoats, uniforms, and the like.

6.1.2 Style A, class 3 machines shall be suitable for stitching, repairing, darning and other operations on uniforms, work clothing, underwear, and general tailor shop use.

6.1.3 Style A, class 4 machines shall be suitable for stitching mackintoshes, bedding, oiled clothing, pile- and sheepskin-lined coats, and light and medium canvas goods.

6.1.4 Style A, class 6 machines shall be suitable for stitching heavy canvas, tents, tarpaulins, sails, awnings, and the like.

6.1.5 Style C, class 1 machines shall be suitable for basting edges and collars of coats, jackets and overcoats, joining pieces of goods together, and other operations.

6.1.6 Style C, class 3 machines shall be suitable for basting edges and collars of coats, jackets, overcoats, and other operations.

6.2 Order data. In addition to ordering data of 6.2 of 00-S-256, the purchaser should include the following information as applicable:

(a) Title, number, and date of this specification.

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(b) Style and class machine required (see 1.2.1).
(c) Specify kinds of feed (see 3.1.1.1, 3.1.2.1, 3.1.3.1 and 3.1.4.1.
(d) Specify working space required (see 3.1.1.1, 3.1.4.1, and 3.2.1.1).
(e) Specify if reversible type feed mechanism shall be furnished (see 3.1.2.1 and 3.1.4.1).
(f) Table top size if other than specified (see 3.1.4.2).

MILITARY CUSTODIANS:

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CIVIL AGENCY INTEREST

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION 00-S-256/1
Sewing Machines, Industrial, Flatbed, Single-Needle (Detail Specification)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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